THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

ANSWER KEY & MARKING SCHEME — MATHS · CLASS 9 · SET 02

Max. Marks: 80 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION A — Objective Type — 20 questions · 20 marks

Q1–Q18 are multiple-choice questions (1 mark each). Q19–Q20 are Assertion (A) and Reason (R) questions: select the correct option from (a), (b), (c), (d) given below each.

1. A point P(a, b) lies in the second quadrant. Then:

(a) $a < 0$ and $b > 0$ (b) $a > 0$ and $b > 0$

(c) $a < 0$ and $b < 0$ (d) $a > 0$ and $b < 0$

Ans: (a) Second quadrant: abscissa negative, ordinate positive.

2. The distance between the points (−2, 5) and (−2, −3) is:

(a) 8 units (b) 5 units (c) √8 units (d) 2 units

Ans: (a) Same x-coordinate, so distance = |5 − (−3)| = 8.

3. The reflection of the point (4, −7) in the y-axis is:

(a) (−4, 7) (b) (4, 7) (c) (4, −7) (d) (−4, −7)

Ans: (d) Reflection in the y-axis changes the sign of the x-coordinate.

4. Which of the following is a linear polynomial?

(a) $\sqrt{x} + 3$ (b) $x^2 - 1$ (c) $2x - 5$ (d) $1/x + 2$

Ans: (c) Degree 1 polynomial in x.

5. The zero of the polynomial p(x) = 5x − 15 is:

(a) 3 (b) −3 (c) 5 (d) 15

Ans: (a) 5x − 15 = 0 ⇒ x = 3.

6. A notebook costs ₹x and a pen costs ₹20. The cost (in ₹) of 6 notebooks and one pen is:

(a) 26x (b) 6x − 20 (c) x + 120 (d) 6x + 20

Ans: (d) 6 × x + 20.

7. Which of the following numbers lies between √2 and √3?

(a) 1.3 (b) 2 (c) 1.5 (d) 1.8

Ans: (c) √2 ≈ 1.414 and √3 ≈ 1.732; only 1.5 lies between them.

8. The decimal expansion of √5 is:

(a) terminating (b) 1.5

(c) non-terminating and non-repeating (d) non-terminating and repeating

Ans: (c) √5 is irrational.

9. If $|x| = \frac{5}{3}$ for a rational number x, then x is:

(a) ± 5/3 (b) 0 (c) −5/3 only (d) 5/3 only

Ans: (a) Both 5/3 and −5/3 have absolute value 5/3.

10. On simplification, (x + 4)(x − 4) is equal to:

(a) $x^2 + 8x - 16$ (b) $x^2 + 16$ (c) $x^2 - 16$ (d) $x^2 - 8x + 16$

Ans: (c) $(a + b)(a - b) = a^2 - b^2$.

11. Using a suitable identity, 51^2 is equal to:

(a) 2651 (b) 2601 (c) 2501 (d) 2611

> **Ans: (b)** $(50 + 1)^2 = 2500 + 100 + 1 = 2601$.

12. Two equal chords AB and CD of a circle with centre O subtend $\angle AOB = 70°$ at the centre. Then $\angle COD$ is:

(a) 110° (b) 140° (c) 70° (d) 35°

> **Ans: (c)** Equal chords subtend equal angles at the centre.

13. The longest chord of a circle of diameter 18 cm has length:

(a) 36 cm (b) $\sqrt{18}$ cm (c) 9 cm (d) 18 cm

> **Ans: (d)** The diameter is the longest chord.

14. An equilateral triangle has perimeter 36 cm. Its area (by Heron's formula) is:

(a) $36\sqrt{3}$ cm² (b) $72\sqrt{3}$ cm² (c) $18\sqrt{3}$ cm² (d) 36 cm²

> **Ans: (a)** Side 12 cm, $s = 18$: $\sqrt{(18 \times 6 \times 6 \times 6)} = 36\sqrt{3}$.

15. The sides of a triangular plot are in the ratio 5 : 12 : 13 and its perimeter is 120 m. The longest side is:

(a) 60 m (b) 20 m (c) 52 m (d) 48 m

> **Ans: (c)** One part = 120 ÷ 30 = 4 m; 13 × 4 = 52 m.

16. Two coins are tossed together. The probability of getting exactly one head is:

(a) 1 (b) $\frac{3}{4}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

> **Ans: (c)** Outcomes HH, HT, TH, TT; exactly one head: HT, TH → 2/4.

17. A letter is picked at random from the word 'PROBABILITY'. The probability of picking the letter B is:

(a) $\frac{4}{11}$ (b) $\frac{2}{11}$ (c) $\frac{1}{11}$ (d) $\frac{3}{11}$

> **Ans: (b)** 11 letters, B appears twice.

18. The 20th term of the sequence 3, 7, 11, 15, … is:

(a) 79 (b) 83 (c) 80 (d) 77

> **Ans: (a)** $t_{20} = 3 + 19 \times 4 = 79$.

19. **Assertion (A):** The perpendicular drawn from the centre of a circle to a chord bisects the chord.
Reason (R): The perpendicular bisector of a chord of a circle passes through the centre of the circle.

> **Ans: (b)** Both are true (a theorem and its converse), but R does not explain A.

20. **Assertion (A):** The probability of an event can be 1.2.
Reason (R): The probability of any event lies between 0 and 1, both inclusive.

> **Ans: (d)** A is false because $P(E) \le 1$; R is a true statement.

SECTION B — Very Short Answer (2 marks each) — 5 questions · 10 marks

Internal choice is provided in one question.

21. The side of a square is $(3x - 2)$ cm. Write the expression for its perimeter and find its value for $x = 4$.

> Perimeter $= 4(3x - 2) =$ **$12x - 8$** cm (1)
> At $x = 4$: $12 \times 4 - 8 =$ **40 cm** (1)

22. Without actual division, state which of $\frac{13}{40}$ and $\frac{7}{30}$ has a terminating decimal expansion. Give a reason.

$40 = 2^3 \times 5$ (only 2 and 5) ⇒ 13/40 terminates (1). $30 = 2 \times 3 \times 5$ has the prime factor 3 ⇒ 7/30 does not terminate (1).

OR

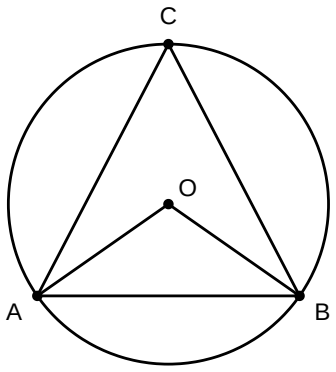
Find two rational numbers between $\frac{2}{3}$ and $\frac{3}{4}$.

2/3 = 32/48 and 3/4 = 36/48 (1). Two rational numbers: **33/48 and 34/48** (= 11/16, 17/24) (1). Any two correct numbers.

23. Evaluate 103×107 using a suitable identity.

$103 \times 107 = (105 - 2)(105 + 2)$ (1) $= 105^2 - 2^2 = 11025 - 4 =$ **11021** (1)

24. In the figure, O is the centre of the circle and ∠ACB = 62°. Find ∠OAB.



∠AOB = 2∠ACB = 124° (1)
OA = OB ⇒ ∠OAB = ∠OBA = (180° − 124°)/2 = **28°** (1)

25. Find the second, third and fourth terms of the sequence given by $t_1 = 2$ and $t_n = (t_{n-1})^2 - 1$ for $n \geq 2$.

$t_2 = 2^2 - 1 = 3$; $t_3 = 3^2 - 1 = 8$; $t_4 = 8^2 - 1 = 63$. (1 for method, 1 for values)

SECTION C — Short Answer (3 marks each) — 6 questions · 18 marks

Internal choice is provided in two questions.

26. Show that $0.428571428571\ldots = \frac{3}{7}$.

Let $x = 0.428571428571\ldots$ Six digits repeat ⇒ $10^6x = 428571.428571\ldots$ (1)
$10^6x - x = 428571$ ⇒ $999999x = 428571$ (1)
$x = 428571/999999 =$ **3/7** (since $428571 = 3 \times 142857$ and $999999 = 7 \times 142857$) (1)

27. In a school, 100 students chose their favourite sport:

Sport	Cricket	Football	Kabaddi	Hockey
Students	40	25	20	15

A student is chosen at random. Find the probability that the student chose (i) kabaddi, (ii) cricket or hockey, (iii) not football.

(i) 20/100 = **1/5** (1) (ii) (40 + 15)/100 = **11/20** (1) (iii) 75/100 = **3/4** (1)

OR

The ages of 100 members of a club were recorded:

Age (years)	11	12	13	14	15
Members	12	18	30	25	15

A member is chosen at random. Find the probability that the member is (i) 13 years old, (ii) older than 13 years, (iii) younger than 12 years.

(i) 30/100 = **3/10** (1) (ii) (25 + 15)/100 = **2/5** (1) (iii) 12/100 = **3/25** (1)

28. A(−3, 2), B(5, 2) and C(5, −4) are three vertices of a rectangle ABCD. Find (i) the coordinates of D, (ii) the length of the diagonal AC, (iii) the area of the rectangle.

(i) D has the x-coordinate of A and the y-coordinate of C: **D(−3, −4)** (1)
(ii) AB = 8, BC = 6 ⇒ AC = √(64 + 36) = **10 units** (1)
(iii) Area = 8 × 6 = **48 square units** (1)

29. Expand: $(2x - 3y + 1)^2$.

Use $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$ with a = 2x, b = −3y, c = 1 (1)
$= 4x^2 + 9y^2 + 1 - 12xy - 6y + 4x$ (2)
= **$4x^2 + 9y^2 + 1 - 12xy + 4x - 6y$**

30. In a circle with centre O, chords AB and CD are equal. Prove that they subtend equal angles at the centre.

In △AOB and △COD: OA = OC, OB = OD (radii) (1); AB = CD (given) (1)
△AOB ≅ △COD (SSS) ⇒ ∠AOB = ∠COD (CPCT) (1)

31. Find the area of the triangle whose sides are 7 cm, 15 cm and 20 cm, using Heron's formula.

s = (7 + 15 + 20)/2 = 21 (1)
Area = √[21 × 14 × 6 × 1] (1) = √1764 = **42 cm²** (1)

O R

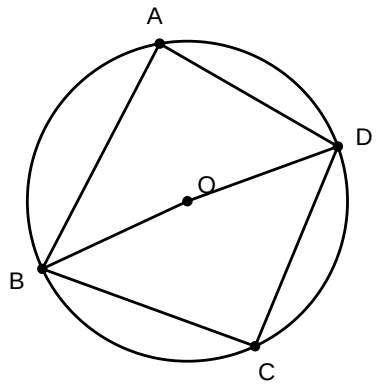
Using Heron's formula, find the area of an equilateral triangle of side 16 cm. (Leave the answer in terms of √3.)

s = 24 (1); Area = √[24 × 8 × 8 × 8] (1) = √12288 = **$64\sqrt{3}$ cm²** (1)

SECTION D — Long Answer (5 marks each) — 4 questions · 20 marks

Internal choice is provided in two questions.

32. Prove that the sum of either pair of opposite angles of a cyclic quadrilateral is 180°.



Given: cyclic quadrilateral ABCD with centre O. To prove: ∠A + ∠C = 180°, ∠B + ∠D = 180°. Construction: join OB and OD. (1)
Arc BCD subtends ∠BOD at the centre and ∠BAD at point A on the remaining circle ⇒ ∠BOD = 2∠BAD (1)
The same arc BAD subtends the reflex angle ∠BOD at the centre and ∠BCD at C ⇒ reflex ∠BOD = 2∠BCD (1)
Adding: ∠BOD + reflex ∠BOD = 2(∠BAD + ∠BCD) (1)
360° = 2(∠A + ∠C) ⇒ ∠A + ∠C = 180°; similarly ∠B + ∠D = 180°. (1)

33. (i) Express 3.2444… (that is, 3.24 with the 4 repeating) in the form p/q in lowest terms. (2½)
(ii) If $x = -\frac{7}{4}$ and $y = \frac{5}{6}$, find |x − y| and |x|·|y|. (2½)

(i) Let x = 3.2444…; 100x = 324.444…, 10x = 32.444… (1) ⇒ 90x = 292 (1) ⇒ x = 292/90 = **146/45** (½)
(ii) x − y = −7/4 − 5/6 = −31/12 ⇒ |x − y| = **31/12** (1½); |x||y| = 7/4 × 5/6 = **35/24** (1)

34. How many two-digit numbers are divisible by 4? Find their sum. (2½ + 2½)

Numbers: 12, 16, …, 96 — AP with a = 12, d = 4, l = 96 (1)
n = (96 − 12)/4 + 1 = **22** (1½)
Sum = 22/2 × (12 + 96) = 11 × 108 = **1188** (2½)

OR

The rungs of a ladder decrease uniformly in length from 45 cm at the bottom to 25 cm at the top. The rungs are 25 cm apart and the top rung is 2½ m above the bottom rung. Find the number of rungs and the total length of wood needed for the rungs.

Number of rungs = 250/25 + 1 = **11** (2)
Lengths form an AP with a = 45, l = 25 (1)
Total length = 11/2 × (45 + 25) = **385 cm** (= 3.85 m) (2)

35. A triangular park has sides 15 m, 20 m and 25 m. Find (i) its area using Heron's formula, (ii) the cost of fencing it at ₹30 per metre, (iii) the cost of turfing it at ₹8 per m². (2 + 1½ + 1½)

(i) s = 30; area = √(30 × 15 × 10 × 5) = √22500 = **150 m²** (2)
(ii) Perimeter = 60 m; cost = 60 × 30 = **₹1800** (1½)
(iii) Cost = 150 × 8 = **₹1200** (1½)

OR

An isosceles triangle has perimeter 36 cm and each of its equal sides is 13 cm. Find the area of the triangle and the length of the altitude on its base.

Base = 36 − 26 = 10 cm (1); s = 18 (½)
Area = √[18 × 5 × 5 × 8] = √3600 = **60 cm²** (2½)
½ × 10 × h = 60 ⇒ h = **12 cm** (1)

SECTION E — Case-Study Based Questions (4 marks each) — 3 questions · 12 marks

Read each case carefully and answer the questions that follow. Internal choice is provided in every 2-mark sub-part.

36. Aarav starts saving for a school trip. He saves ₹100 in the first month and increases his saving by ₹50 every month, so his saving in the n-th month is $t_n = 50n + 50$.

(i) How much does he save in the 6th month?

$t_6 = 50 \times 6 + 50 =$ **₹350**

(ii) The monthly savings form an AP. Write its common difference.

$t_2 - t_1 = 150 - 100 =$ **₹50**

(iii) In which month will he save ₹1000?

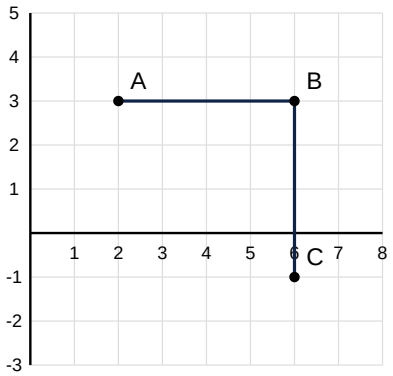
50n + 50 = 1000 (1) ⇒ n = **19th month** (1)

OR

(iii) Find the total amount he saves in the first 10 months.

a = 100, l = t_{10} = 550 (1); S = 10/2 × (100 + 550) = **₹3250** (1)

37. The map of a colony is drawn on a grid (1 unit = 100 m). Meera's house is at A(2, 3), Karan's house is at B(6, 3) and Zoya's house is at C(6, −1).



(i) Find the distance between Meera's and Karan's houses (in units).

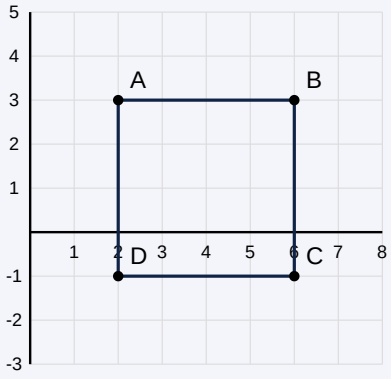
AB = |6 − 2| = **4 units** (= 400 m)

(ii) The school is at the midpoint of AC. Find its coordinates.

$(\frac{2+6}{2}, \frac{3-1}{2})$ = **(4, 1)**

(iii) A park is to be located at D so that ABCD is a square. Find the coordinates of D.

AB = BC = 4 and angle B is a right angle (1) ⇒ D(2, −1) (1)



OR

(iii) Find the straight-line distance between the houses of Meera and Zoya.

AC = √[(6 − 2)² + (−1 − 3)²] (1) = √32 = **4√2 units** (1)

38. A spinner is divided into 8 equal sectors numbered 1 to 8. The spinner is spun once and the number on the sector that stops at the pointer is noted.

(i) Find the probability of getting the number 7.

P = **1/8**

(ii) Find the probability of getting a prime number.

Primes: 2, 3, 5, 7 ⇒ P = 4/8 = **1/2**

(iii) Find the probability of getting a number greater than 5.

Favourable: 6, 7, 8 (1) ⇒ P = **3/8** (1)

OR

(iii) Find the probability of getting a multiple of 3 or a multiple of 4.

Favourable: 3, 4, 6, 8 (1) ⇒ P = 4/8 = **1/2** (1)